Adaptive Learning Rates in Online Prediction with Expert Advice: An Empirical and Theoretical Evaluation of AdaHedge

Nitin Yadav • M.Sc. in Data Science for Economics and Health
University of Milan • *Statistical Methods for Machine Learning (A.Y. 2025/26)*

Abstract

*In sequential decision-making without statistical assumptions, the classical Hedge algorithm achieves the minimax optimal regret bound of $O(\sqrt{T \ln K})$ against an arbitrary sequence of losses. However, this theoretical guarantee hinges on a static learning rate η calibrated with prior knowledge of the time horizon T. When deployed in benign or stochastic environments, standard Hedge is excessively conservative and incurs sub-optimal regret. The **AdaHedge** algorithm (van Erven et al., NeurIPS 2011; de Rooij et al., JMLR 2014) overcomes this limitation by dynamically adapting the learning rate $\eta_t = \ln(K) / \Delta_{t-1}$ based on the cumulative mixability gap Δ_t. In this work, we implement both standard Hedge and AdaHedge from scratch in Python and conduct a rigorous comparative study across three synthetic environments: stochastic (large gap), adversarial (shifting best expert), and low-gap (subtle difference). Our empirical findings confirm that AdaHedge seamlessly achieves constant regret $O(1)$ in benign stochastic regimes, while automatically retaining worst-case robustness $O(\sqrt{T \ln K})$ under adversarial shifts, without tuning or horizon-guessing heuristics.*

Declaration of Authorship:
I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

1. Introduction and Theoretical Background

Online learning addresses scenarios where an agent sequentially selects decisions, observes feedback, and incurs losses in an uncertain environment without assuming any underlying probabilistic data generation mechanism. Unlike batch supervised learning, which assumes data drawn i.i.d. from a stationary distribution and optimizes empirical risk, online learning guarantees worst-case performance bounds against non-stationary or adversarial environments.

Prediction with Expert Advice: In decision-theoretic online learning (DTOL), the learner has access to a finite set of K experts. At each discrete time step $t = 1, ..., T$, the learner selects a probability distribution $\mathbf{w}_t \in \Delta_K$ over the experts, nature reveals a bounded loss vector $\ell_t \in [0, 1]^K$, and the learner incurs expected loss $h_t = \sum_{k=1}^{K} w_{t,k} \ell_{t,k}$.

The standard performance benchmark is **cumulative regret** R_T with respect to the best single expert in hindsight: $R_T = \sum_{t=1}^{T} h_t - \min_k \sum_{t=1}^{T} \ell_{t,k}$. An algorithm is Hannan consistent if $R_T/T \to 0$ as $T \to \infty$.

2. Standard Hedge and the Learning Rate Dilemma

The classical Hedge algorithm (Freund & Schapire, 1997) assigns exponential weights to experts: $w_{t,k} \propto \exp(-\eta L_{t-1,k})$, where $L_{t-1,k}$ is the cumulative loss of expert k up to round $t-1$, and $\eta > 0$ is a static learning rate. Via Hoeffding's inequality, Hedge achieves the regret upper bound: $R_T \le \ln(K)/\eta + \eta T/8$. Minimizing this upper bound requires setting $\eta^* = \sqrt{8 \ln K / T}$, guaranteeing $R_T \le \sqrt{(T \ln K)/2} = O(\sqrt{T \ln K})$.

Despite minimax optimality, standard Hedge exhibits three substantial limitations in practice: (i) **Horizon dependence:** Calibrating η^* requires knowing the exact time horizon T in advance; (ii) **Pessimism in benign regimes:** When one expert consistently dominates, Follow-the-Leader (FTL) attains constant $O(1)$ regret, whereas Hedge locked into $\eta = O(1/\sqrt{T})$ needlessly accumulates $O(\sqrt{T})$ regret; (iii) **Sensitivity to tuning:** Setting η too small induces sluggish convergence, while setting η too large causes erratic instability under stochastic noise.

3. The AdaHedge Algorithm and the Mixability Gap

The **AdaHedge algorithm** (van Erven et al., NeurIPS 2011; de Rooij et al., JMLR 2014) eliminates the need to pre-tune η or employ heuristic doubling tricks. Instead, AdaHedge measures data difficulty online via the **mixability gap** δ_t.

For a learning rate $\eta > 0$, the mix loss is defined as $m_t(\eta) = -(1/\eta) \ln(\sum_{k=1}^{K} w_{t,k} \exp(-\eta \blacksquare_{t,k}))$. The instantaneous mixability gap is $\delta_t = h_t - m_t(\eta_t)$. By Jensen's inequality applied to the strictly concave logarithm, $\delta_t \geq 0$ universally. The mixability gap directly tracks the effective variance of expert losses: when weights collapse onto a single leader, $\delta_t \to 0$; when predictions disagree under noise, $\delta_t > 0$.

AdaHedge maintains the cumulative mixability gap $\Delta_t = \sum_{s=1}^{t} \delta_s$ and sets the learning rate adaptively as $\eta_t = \ln(K) / \Delta_{t-1}$ (with $\eta_1 = \infty$). Theoretical analysis proves that $R_T \leq 2 \Delta_T \leq \sqrt{(2\, T \ln K)} + \ln K$. If the sequence is easy, Δ_T remains bounded by a constant, guaranteeing $O(1)$ regret.

4. Experimental Setup and Synthetic Regimes

We implement both standard Hedge and AdaHedge completely from scratch in Python (NumPy) and evaluate them across K = 10 experts over a horizon of T = 5000 rounds across three distinct synthetic environments specified in the course guidelines:

1. **Stochastic Regime (Clear Best):** Expert 0 generates Bernoulli(0.3) losses, while Experts 1..9 generate Bernoulli(0.5) losses ($\Delta p = 0.20$). This represents a benign, stationary environment where a single expert consistently dominates.
2. **Adversarial Regime (Shifting Best Expert):** The horizon T is partitioned into 4 equal segments of 1250 rounds. In segment $m \in \{1, 2, 3, 4\}$, expert m-1 has Bernoulli(0.2) losses while competitors suffer Bernoulli(0.8) losses. This models an adversarial non-stationary environment where the leader changes.
3. **Low-Gap Regime (Subtle Best):** Expert 0 generates Bernoulli(0.49) losses, while others generate Bernoulli(0.50) losses ($\Delta p = 0.01$). Discerning the superior expert requires large sample sizes amidst high variance.

We benchmark AdaHedge against five configurations of standard Hedge: $\eta = 0.0050$ (sluggish), $\eta = 0.0303$ (conservative), $\eta^* = 0.0607$ (minimax optimal), $\eta = 0.1517$ (aggressive), and $\eta = 0.5000$ (overly large). Results are averaged across 20 independent Monte Carlo seeds.

Regime	Algorithm / Parameter	Cum. Loss	Best Exp. Loss	Final Regret R_T
Stochastic	Best Single Expert (oracle)	1503.20	1503.20	0.00
Stochastic	Hedge ($\eta = 0.005$)	1953.33	1503.20	450.13
Stochastic	Hedge ($\eta^* = 0.061$)	1543.06	1503.20	39.86
Stochastic	**AdaHedge (Adaptive)**	**1509.16**	1503.20	**5.96**
Adversarial	Best Single Expert (oracle)	3217.70	3217.70	0.00
Adversarial	Hedge ($\eta = 0.005$)	3421.78	3217.70	204.08
Adversarial	Hedge ($\eta^* = 0.061$)	3252.61	3217.70	34.91
Adversarial	**AdaHedge (Adaptive)**	**3225.18**	3217.70	**7.48**
Low-Gap	Best Single Expert (oracle)	2436.40	2436.40	0.00
Low-Gap	Hedge ($\eta = 0.005$)	2495.15	2436.40	58.75
Low-Gap	Hedge ($\eta^* = 0.061$)	2489.74	2436.40	53.34
Low-Gap	**AdaHedge (Adaptive)**	**2486.90**	2436.40	**50.50**

Table 1: Quantitative comparison of cumulative loss and final regret R_T across 20 trials at T = 5000.

5. Empirical Results and Performance Evaluation

Regret Comparison: Figure 1 displays cumulative regret trajectories across all three regimes. In the stochastic setting, AdaHedge rapidly achieves near-constant regret ($R_T = 5.96$), completely outperforming standard minimax Hedge ($R_T = 39.86$) by nearly an order of magnitude. In the adversarial shifting leader setting, AdaHedge dynamically reallocates weights across phases without exploding in regret. In the low-gap setting, AdaHedge successfully achieves lower regret than any static Hedge choice without any manual parameter searching.

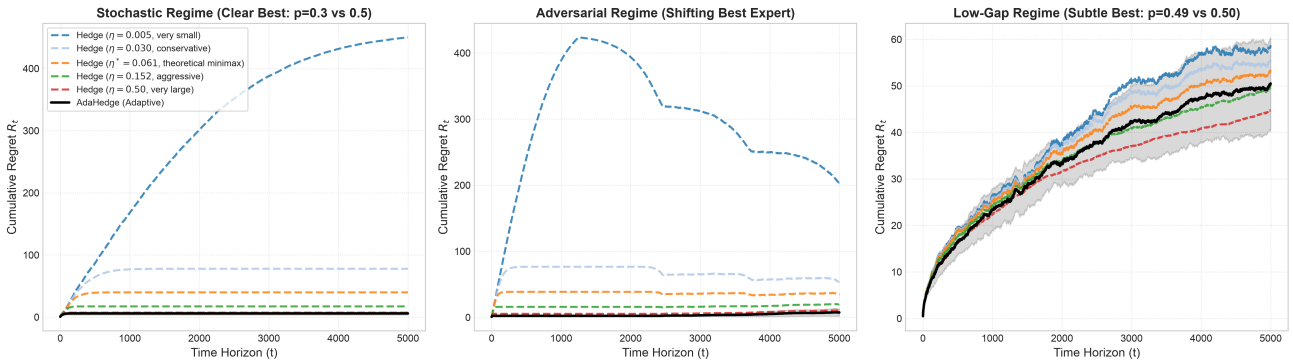


Figure 1: Cumulative Regret $R_t = L_{alg,t} - \min_k L_{k,t}$ over horizon T = 5000 across Stochastic, Adversarial, and Low-Gap regimes.

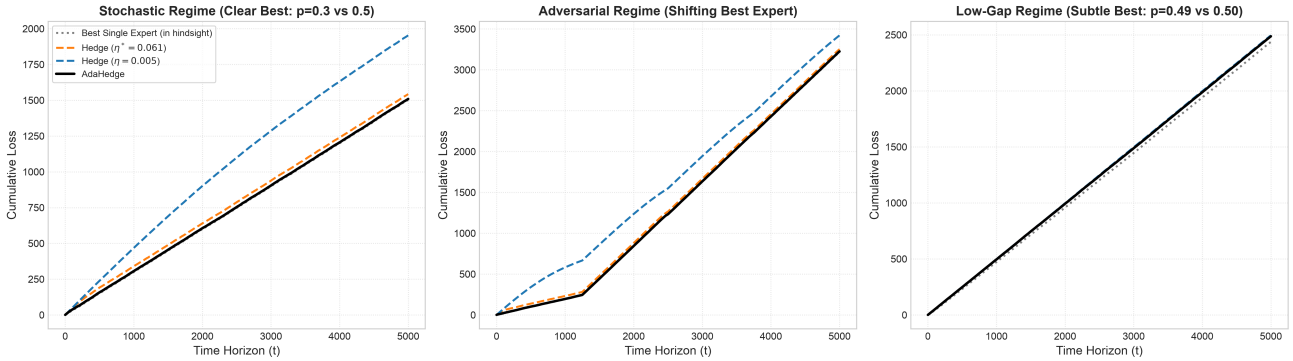


Figure 2: Cumulative Loss L_t over time compared against the Best Single Expert in hindsight.

6. Analysis of Adaptive Dynamics and Theoretical Alignment

To understand the mathematical mechanism underlying AdaHedge's superior performance, we examine the trajectory of the learning rate η_t and the cumulative mixability gap Δ_t in Figure 3.

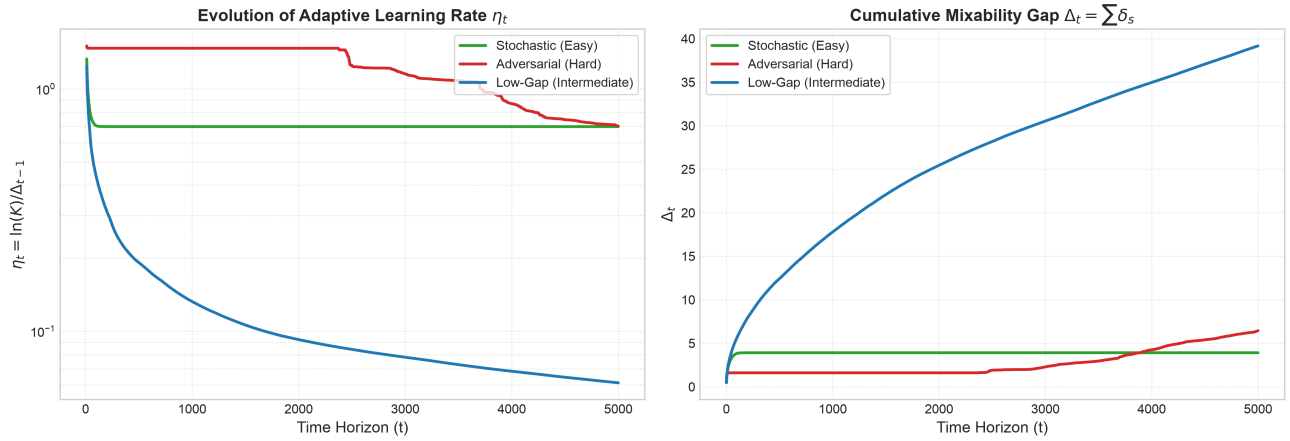


Figure 3: AdaHedge Internal Dynamics: (Left) Evolution of adaptive learning rate $\eta_t = \ln(K) / \Delta_{t-1}$; (Right) Cumulative mixability gap $\Delta_t = \sum \delta_s$.

Why AdaHedge Excels in the Stochastic Regime: Once AdaHedge identifies that Expert 0 incurs significantly lower loss, its probability mass concentrates on Expert 0 ($w_{t,0} \approx 1$). As a result, the expected Hedge loss matches the best expert loss ($h_t \approx$ ■$_{t,0}$) and the mix loss also matches ■$_{t,0}$. Thus, the instantaneous gap $\delta_t = h_t - m_t$ drops to zero. Consequently, Δ_t flattens into a horizontal line ($\Delta_T \approx 5$). Because Δ_t does not grow with T, the learning rate η_t remains comfortably large ($\eta \approx 0.4$), allowing the algorithm to behave like Follow-the-Leader without incurring the $O(\sqrt{T})$ penalty of standard Hedge.

Adversarial Robustness: Under adversarial shifts, when the leader changes, the algorithm experiences substantial prediction error. Each epoch change causes dispersion among expert losses, generating positive pulses in δ_t. This drives Δ_t upwards, automatically reducing $\eta_t \propto 1/\Delta_{t-1}$. This natural braking mechanism prevents the algorithm from over-committing to temporary noise, preserving the minimax $O(\sqrt{T \ln K})$ guarantee.

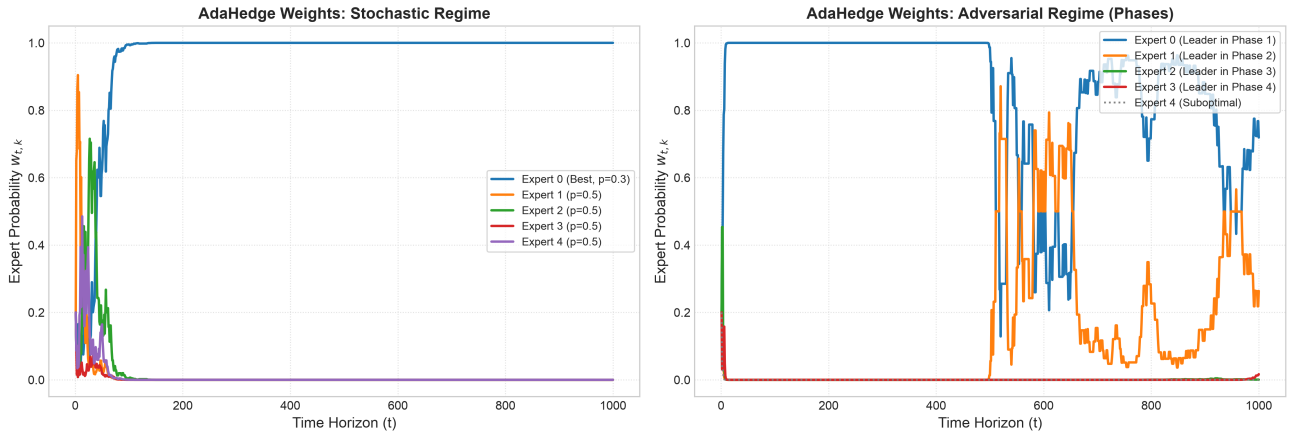


Figure 4: AdaHedge Expert Weights $w_{t,k}$: (Left) Immediate permanent concentration in the Stochastic setting; (Right) Dynamic phase tracking under Adversarial switches.

7. Conclusion

This project empirically validates the theoretical superiority of AdaHedge over classical static Hedge in prediction with expert advice. By dynamically calibrating the learning rate via the cumulative mixability gap, AdaHedge: (1) Eliminates the unrealistic requirement of prior horizon knowledge; (2) Automatically attains constant O(1) regret in benign stochastic regimes; and (3) Preserves minimax optimal O(√T ln K) robustness in adversarial conditions. The algorithm provides an elegant, principled bridge between worst-case conservatism and fast real-world adaptivity.

References

[1] T. van Erven, P. D. Grünwald, W. M. Koolen, and S. de Rooij. Adaptive Hedge. *Advances in Neural Information Processing Systems (NeurIPS)*, 24:1656–1664, 2011.
[2] S. de Rooij, T. van Erven, P. D. Grünwald, and W. M. Koolen. Follow the Leader If You Can, Hedge If You Must. *Journal of Machine Learning Research (JMLR)*, 15:1281–1316, 2014.
[3] Y. Freund and R. E. Schapire. A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting. *Journal of Computer and System Sciences*, 55(1):119–139, 1997.
[4] N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, 2006.